

- Operate 24/7 with no downtime
- Scale applications and adapt to increased demand
- Maintain consistent performance

Types of AI agents

There are several types of AI agents, and each is designed for specific tasks and environments.

Simple reflex agents: These agents make decisions based on the current perception. They follow condition-action rules and are suitable for observable environments. They have no learning skills or memory. Automated hand sanitiser dispensers are an example. These devices dispense the sanitiser when they detect motion, ensuring hygiene without human intervention.

Model-based reflex agents: These agents handle partially observable environments by maintaining an internal state based on perception history. Smart IV pumps are an example. These pumps adjust the flow rate of intravenous fluids based on real-time patient data, such as heart rate and blood pressure, ensuring safe and precise delivery.

Goal-based agents: These agents have goals and choose actions to achieve them. They consider the current state and the desired goal to make decisions. They are centred not only on existing but also on future conditions, and on

the relationship between conditions and operations. Personalised treatment planning systems in healthcare are an example. These systems analyse patient data, including genetics and medical history, to create customised treatment plans for managing chronic diseases.

Utility-based agents: These choose actions that maximise the expected utility. An example is resource allocation systems in hospitals. These systems optimise the use of hospital resources, such as beds and medical equipment, to maximise patient care efficiency and minimise costs.

Learning agents: These agents learn from their experiences and improve their performance over time. They start with basic knowledge and adapt through learning. For example, in predictive analytics for patient monitoring, these agents learn from historical patient data to predict potential health issues, allowing for early intervention and better patient outcomes.

Hierarchical agents: These agents decompose tasks into subtasks and solve them hierarchically. They can handle complex tasks by breaking them down into simpler, manageable parts. Hospital workflow management systems are an example. These systems break down complex tasks into subtasks, such as patient admission, treatment scheduling, and discharge planning, to streamline hospital operations.

Key characteristics of AI agents

AI agent systems possess several key characteristics that distinguish them from traditional AI systems.

Autonomy: AI agent systems operate independently. They can make decisions and execute actions without human intervention.

Adaptability: By learning from experiences, AI agents can adapt to new environments and scenarios.

Goal-driven: They are designed to achieve specific objectives or goals.

Context awareness: Agent systems can understand and respond to the context in which they operate, allowing for more nuanced and effective interactions.

Proactivity: They can anticipate future needs and take proactive measures to address them, rather than simply reacting to external inputs.

Technical capabilities: These systems can leverage a variety of tools to enhance their functionality and handle complex scenarios.

- **Internet access:** Allows agents to retrieve real-time information, perform web searches, and gather data from online sources.
- **Code interpreters:** Enable agents to execute and interpret code, facilitating complex computations, data analysis, and automation tasks.
- **API calls:** Allow agents to interact with external services and systems, enabling seamless integration with various platforms, databases, and applications.

Communication: Effective communication is a hallmark of an AI agent, enabling them to interact seamlessly with humans and other AI systems.

Scalability: AI agents are designed to scale efficiently, handling increased workloads and complexity as needed.

Robustness: AI agent systems are resilient and can maintain performance even in the face of challenges or disruptions.

Ethical considerations: These systems are built with ethical guidelines



Figure 1: Types of AI agents